Primordial Black Holes and the Early Universe

Seeds of the First Structures

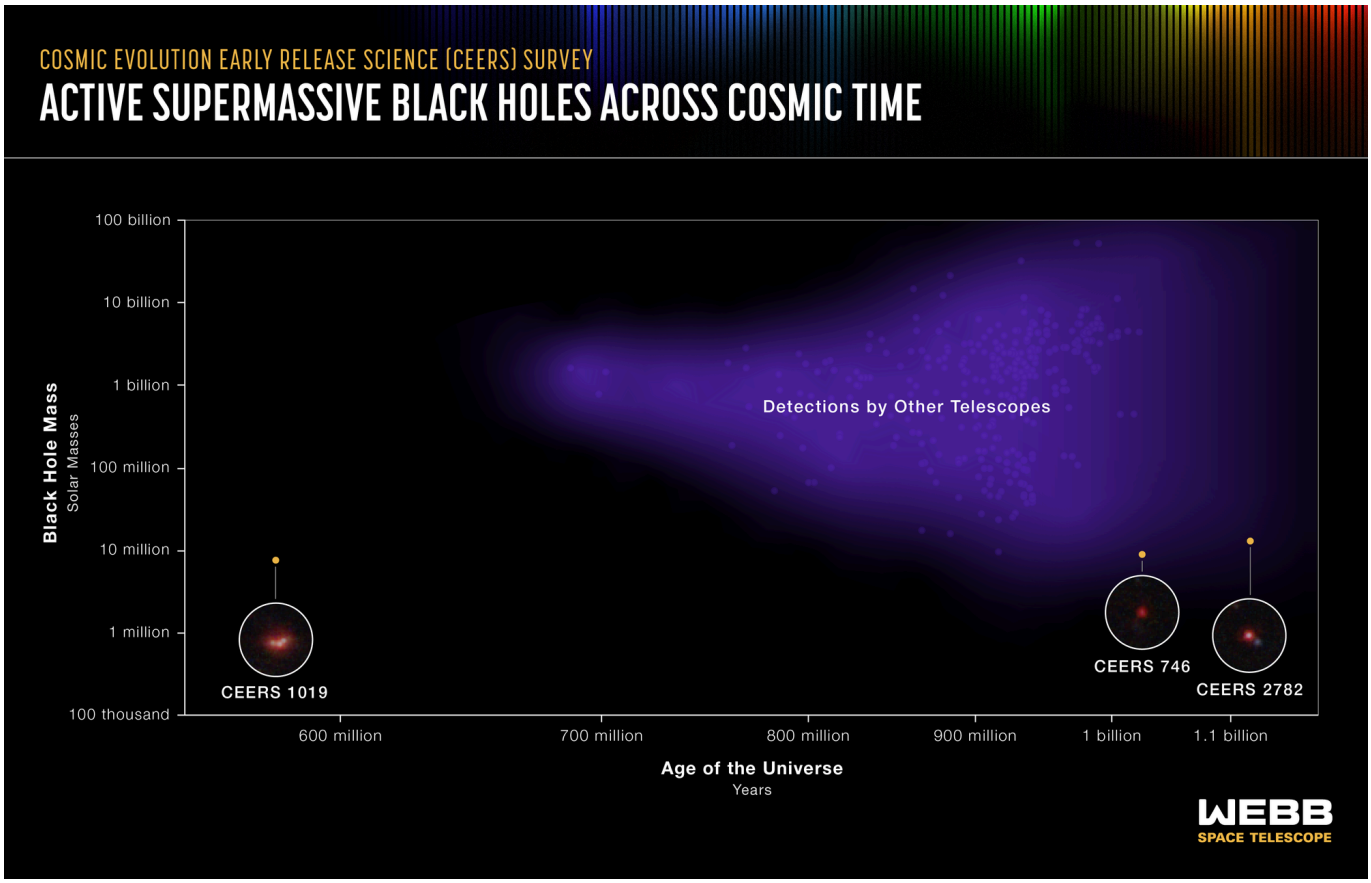
Primordial Black Holes (PBHs) likely formed in high-density regions of the early universe. These seeds acted as gravitational anchors, helping to cluster dark matter into the halos that eventually allowed the first stars and galaxies to form.

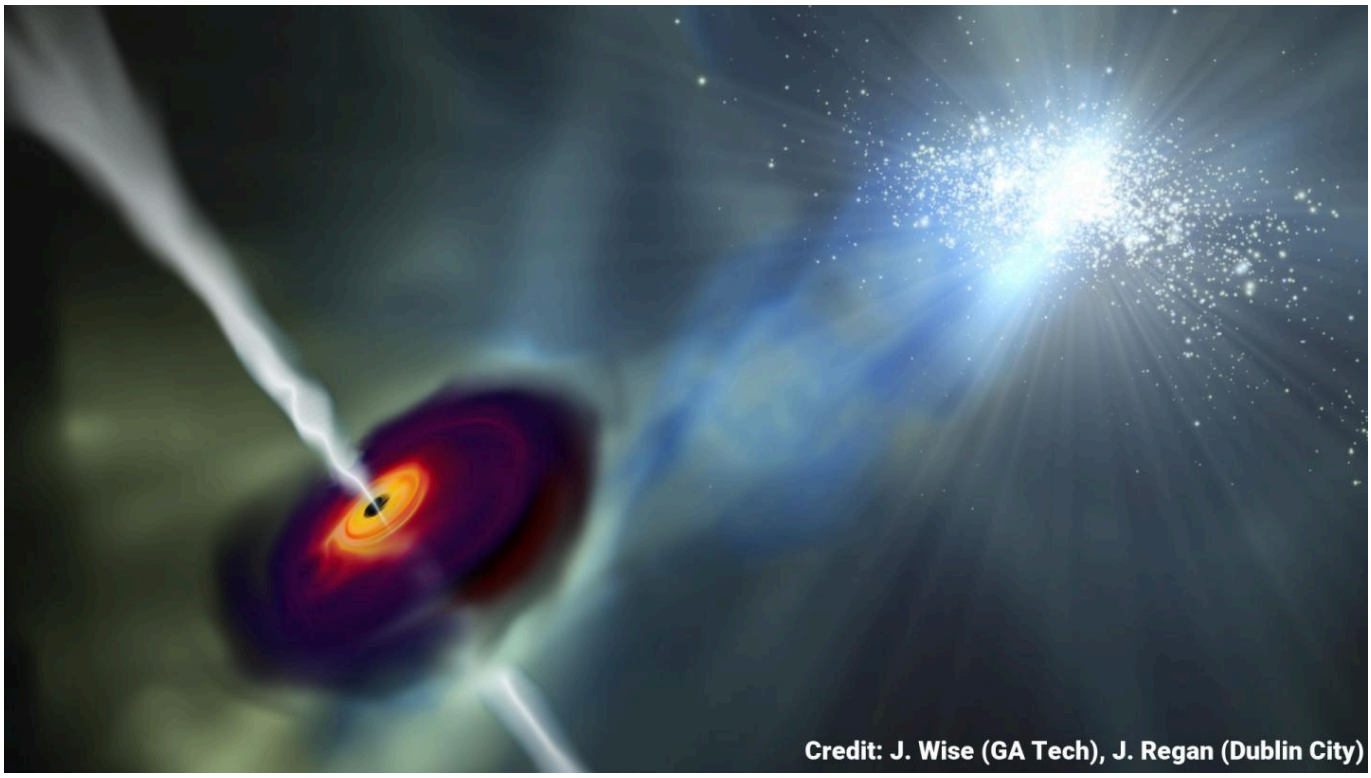
The Supermassive Mystery

We see "impossible" supermassive black holes—billions of times the mass of our sun—existing less than a billion years after the Big Bang. Massive PBHs (over 1,000 solar masses) provide a shortcut, acting as the "missing link" that allowed these giants to grow so quickly in the infant universe.

The Matter Mystery: Baryogenesis

Why is there anything in the universe? PBHs may be linked to baryogenesis– the process that allowed matter to triumph over antimatter. Some theories suggest the violent explosions of tiny black holes helped shape the very matter that makes up everything we see today.


COSMIC EVOLUTION EARLY RELEASE SCIENCE (CEERS) SURVEY
ACTIVE SUPERMASSIVE BLACK HOLES ACROSS COSMIC TIME
Black Hole Mass
Solar Masses
100 billion
10 billion
1 billion
100 million
10 million
1 million
100 thousand
Detections by Other Telescopes
CEERS 1019
CEERS 746
CEERS 2782
600 million
700 million
800 million
900 million
1 billion
1.1 billion
Age of the Universe
Years
WEBB
SPACE TELESCOPE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